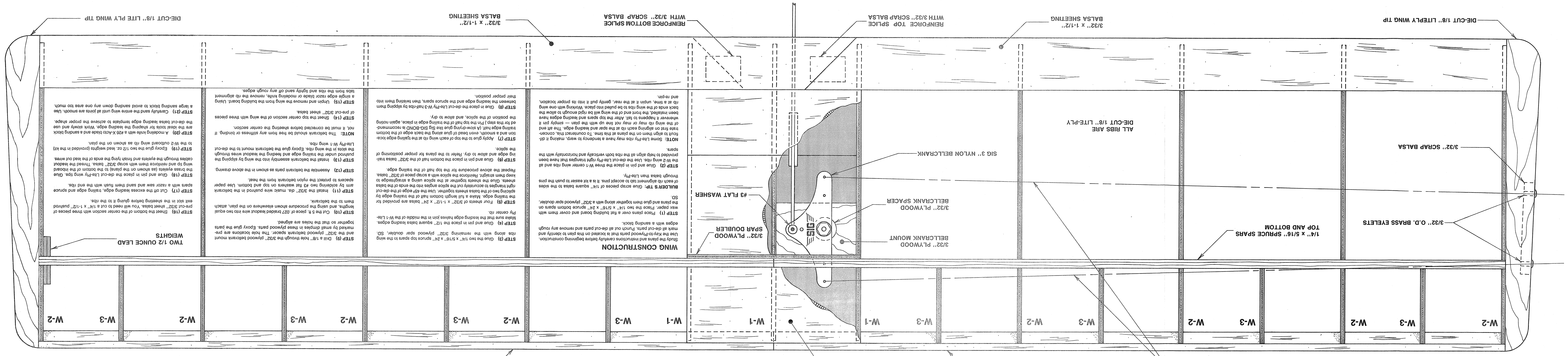


DIE-CUT 1/8" LITE PLY WING TIP
3/32" x 1-1/2" BALSA SHEETING
REINFORCE BOTTOM SPLICE WITH 3/32" SCRAP BALSA
REINFORCE TOP SPLICE WITH 3/32" SCRAP BALSA
3/32" x 1-1/2" BALSA SHEETING
DIE-CUT 1/8" LITEPLY WING TIP
TWO 1/2 OUNCE LEAD WEIGHTS
ALL RIBS ARE DIE-CUT 1/8" LITE-PLY
3/32" SCRAP BALSA
3/32" O.D. BRASS EYELETS
1/4" x 5/16" SPRUCE SPARS TOP AND BOTTOM
SIG 3" NYLON BELLCRANK
3/32" PLYWOOD BELLCRANK SPACER
3/32" PLYWOOD BELLCRANK MOUNT
#3 FLAT WASHER
3/32" PLYWOOD SPAR DOUBLER
SIG
W-1
W-2
W-3
1/2" SQUARE BALSA LEADING EDGE
3/32" BALSA SHEETING
1/2" SQUARE BALSA LEADING EDGE
.027 BRAIDED LEADOUT WIRE

WING CONSTRUCTION

Study the plans and instructions carefully before beginning construction. Use the Key-to-Plywood parts that is located on this plan to identify and mark all die-cut parts. Punch out all die-cut parts and remove any rough edges with a sanding block.

STEP (1) Place plans over a flat building board and cover them with wax paper. Place the two 1/4" x 5/16" x 24" spruce bottom spars on the plans and glue them together along with a 3/32" plywood spar doubler, SD.

BUILDER'S TIP: Glue scrap pieces of 1/4" square balsa to the sides of each rib alignment tab to accept pins. It is a lot easier to push the pins through balsa than Lite-Ply.

STEP (2) Glue and pin in place the three W-1 center wing ribs and all the W-2 wing ribs. Use the die-cut Lite-Ply right triangles that have been provided to help align all the ribs both vertically and horizontally with the spars.

NOTE: Some Lite-Ply ribs may have a tendency to warp, making it difficult to align them on the plans at this time. To counteract this, concentrate first on aligning each rib at the spar and leading edge. The aft end of the wing rib may or may not line up with the plan — simply pin it wherever it happens to fall. After the top spars and leading edges have been installed, the front end of the wing will be rigid enough to allow the back ends of the wing ribs to be pulled into place. Working with one wing rib at a time, unpin it at the rear, gently pull it into its proper location, and re-pin.

STEP (3) Glue the two 1/4" x 5/16" x 24" spruce top spars to the wing ribs along with the remaining 3/32" plywood spar doubler, SD.

STEP (4) Glue and pin in place the 1/2" square balsa leading edges. Make sure that the leading edge halves join in the middle of the W-1 Lite-Ply center rib.

STEP (5) Four sheets of 3/32" x 1-1/2" x 24" balsa are provided for the trailing edge. Make a full length bottom half of the trailing edge by splicing two of the balsa sheets together. Use the 45⁰ edge of the die-cut right triangles to accurately cut the splice angles into the ends of the balsa sheets. Glue the sheets together at the splice using a straightedge to keep them straight. Reinforce the splice with a scrap piece of 3/32" balsa. Repeat the above procedure for the top half of the trailing edge.

STEP (6) Glue and pin in place the bottom half of the 3/32" balsa trailing edge and allow to dry. Refer to the plans for proper positioning of the splice.

STEP (7) Apply glue to the top of each wing rib at the tgailing edge location and a smooth, even bead of glue along the back edge of the bottom trailing edge half. (A slow-drying glue like Sig SIG-BOND is recommended for this step.) Pin the top half of the trailing edge in place, again noting the position of the splice, and allow to dry.

STEP (8) Glue in place the die-cut Lite-Ply W-3 half-ribs by slipping them between the leading edge and the spruce spars, then twisting them into their proper position.

STEP (9) Drill a 1/8" hole through the 3/32" plywood bellcrank mount and the 3/32" plywood bellcrank spacer. The hole locations are premarked by small dimples in these plywood parts. Epoxy glue the parts together so that the holes are aligned.

STEP (10) Cut the 5 ft. piece of .027 braided leadout wire into two equal lengths, and using the procedure shown elsewhere on the plan, attach them to the bellcrank.

STEP (11) Install the 3/32" dia. music wire pushrod in the bellcrank arm by soldering two #3 flat washers on top and bottom. Use paper spacers to protect the nylon bellcrank from the heat.

STEP (12) Assemble the bellcrank parts as shown in the above drawing.

STEP (13) Install the bellcrank assembly into the wing by slipping the pushrod under the trailing edge and feeding the leadout wires through the slots in the wing ribs. Epoxy glue the bellcrank mount to the die-cut Lite-Ply W-1 wing ribs.

NOTE: The bellcrank should be free from any stiffness or binding. If not, it must be corrected before sheeting the center section.

STEP (14) Sheet the top center section of the wing with three pieces of pre-cut 3/32" sheet balsa.

STEP (15) Unpin and remove the wing from the building board. Using a single edge razor blade or modeling knife, remove the rib alignment tabs from the ribs and lightly sand off any rough edges.

STEP (16) Sheet the bottom of the center section with three pieces of pre-cut 3/32" sheet balsa. You will need to cut a 1/4" x 1-1/2" pushrod exit slot in the sheeting before gluing it to the ribs.

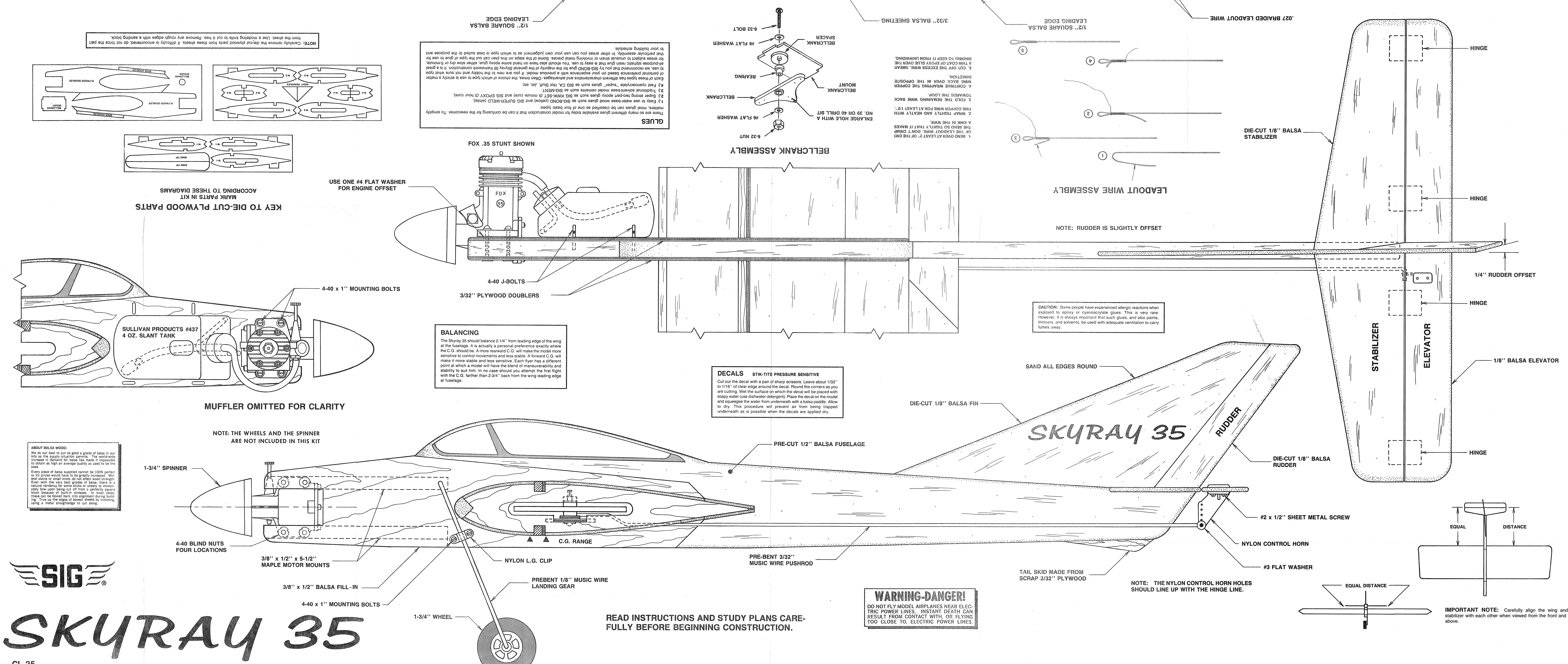
STEP (17) Cut off the excess leading edge, trailing edge and spruce spars with a razor saw and sand them flush with the end ribs.

STEP (18) Glue and pin in place the die-cut Lite-Ply wing tips. Glue the brass eyelets (as shown on the plans) to the bottom of the inboard wing tip and reinforce them with scrap 3/32" balsa. Thread the leadout cables through the eyelets and finish tying the ends of the lead out wires.

STEP (19) Epoxy glue the two 1/2 oz. lead weights (provided in the kit) to the W-2 outboard wing rib as shown on the plan.

STEP (20) A modeling knife with a #26 X-Acto blade and a sanding block are the ideal tools for shaping the leading edge. Work slowly and use the die-cut balsa leading edge template to achieve the proper shape.

STEP (21) Carefully sand the entire wing until all joints are smooth. Use a large sanding block to avoid sanding down any one area too much.



KEY TO DIE-CUT PLYWOOD PARTS
MARK PARTS IN KIT ACCORDING TO THESE DIAGRAMS
PLYWOOD DOUBLER
SPAR DOUBLER
BELLCRANK SPACER
BELLCRANK MOUNT
RIGHT TRIANGLES
WING TIP
W-2
W-3
BELLCRANK ASSEMBLY
6-32 NUT
#6 FLAT WASHER
BELLCRANK
BEARING
BELLCRANK MOUNT
BELLCRANK SPACER
#6 FLAT WASHER
6-32 BOLT
ENLARGE HOLE WITH A NO. 39 OR 40 DRILL BIT
LEADOUT WIRE ASSEMBLY
1
2
3
4
5
FOX .35 STUNT SHOWN
USE ONE #4 FLAT WASHER FOR ENGINE OFFSET
FOX
4-40 J-BOLTS
3/32" PLYWOOD DOUBLERS
NOTE: RUDDER IS SLIGHTLY OFFSET
DIE-CUT 1/8" BALSA STABILIZER
HINGE
HINGE
HINGE
HINGE
1/4" RUDDER OFFSET
STABILIZER
ELEVATOR
1/8" BALSA ELEVATOR
4-40 x 1" MOUNTING BOLTS
SULLIVAN PRODUCTS #437 4 OZ. SLANT TANK
MUFFLER OMITTED FOR CLARITY
NOTE: THE WHEELS AND THE SPINNER ARE NOT INCLUDED IN THIS KIT
1-3/4" SPINNER
4-40 BLIND NUTS FOUR LOCATIONS
3/8" x 1/2" x 5-1/2" MAPLE MOTOR MOUNTS
3/8" x 1/2" BALSA FILL-IN
4-40 x 1" MOUNTING BOLTS
NYLON L.G. CLIP
PREBENT 1/8" MUSIC WIRE LANDING GEAR
1-3/4" WHEEL
C.G. RANGE
PRE-CUT 1/2" BALSA FUSELAGE
PRE-BENT 3/32" MUSIC WIRE PUSHROD
SAND ALL EDGES ROUND
DIE-CUT 1/8" BALSA FIN
SKYRAY 35
RUDDER
DIE-CUT 1/8" BALSA RUDDER
#2 x 1/2" SHEET METAL SCREW
NYLON CONTROL HORN
#3 FLAT WASHER
TAIL SKID MADE FROM SCRAP 3/32" PLYWOOD
NOTE: THE NYLON CONTROL HORN HOLES SHOULD LINE UP WITH THE HINGE LINE.
EQUAL
DISTANCE
EQUAL DISTANCE

NOTE: Carefully remove the die-cut plywood parts from these sheets. If difficulty is encountered, do not force the part from the sheet. Use a modeling knife to cut it free. Remove any rough edges with a sanding block.

1. BEND OVER AT LEAST 2" OF THE END OF THE LEADOUT WIRE. DON'T CRIMP THE BEND SO TIGHTLY THAT IT MAKES A KINK IN THE WIRE.
2. WRAP TIGHTLY AND NEATLY WITH FINE COPPER WIRE FOR AT LEAST 1/2".
3. FOLD THE REMAINING WIRE BACK TOWARDS THE LOOP.
4. CONTINUE WRAPPING THE COPPER WIRE BACK OVER IN THE OPPOSITE DIRECTION.
5. CUT OFF THE EXCESS WIRE. SMEAR A THIN COAT OF EPOXY GLUE OVER THE BINDING TO KEEP IT FROM UNWINDING.

GLUES

There are so many different glues available today for model construction that it can be confusing for the newcomer. To simplify matters, most glues can be classified as one of four basic types:

1.) Easy to use water-base wood glues such as SIG-BOND (yellow) and SIG SUPER-WELD (white).
2.) Super strong two-part epoxy glues such as SIG KWIK-SET (5 minute cure) and SIG EPOXY (3 hour cure).
3.) Traditional solvent-base model cements such as SIG-MENT.
4.) Fast cyanoacrylate "super" glues such as SIG CA, Hot Stuff, Jet, etc.

Each of these types has different characteristics and advantages. Often times, the choice of which type to use is strictly a matter of personal preference based on your experience with a previous model. If you are new to the hobby and not sure what type to use, we recomment that you try SIG-BOND glue for the majority of the general Skyray 35 framework construction. It is a great all-purpose alphatic resin glue that is easy to use. You should also have on hand some epoxy glue, either slow dry or 5-minute, for areas subject to unusual strain or involving metal pieces. Some of the steps on this plan call out the type of glue to use for that particular assembly. In other areas you can use your own judgement as to which type is best suited to the purpose and to your building schedule.

BALANCING

The Skyray 35 should balance 2-1/4" from leading edge of the wing at the fuselage. It is actually a personal preference exactly where the C.G. should be. A more rearward C.G. will make the model more sensitive to control movements and less stable. A forward C.G. will make it more stable and less sensitive. Each flyer has a different point at which a model will have the blend of maneuverability and stability to suit him. In no case should you attempt the first flight with the C.G. farther than 2-3/4" back from the wing leading edge at fuselage.

DECALS STIK-TITE PRESSURE SENSITIVE

Cut out the decal with a pair of sharp scissors. Leave about 1/32" to 1/16" of clear edge around the decal. Round the corners as you are cutting. Wet the surface on which the decal will be placed with soapy water (use dishwater detergent). Place the decal on the model and squeegee the water from underneath with a balsa paddle. Allow to dry. This procedure will prevent air from being trapped underneath as is possible when the decals are applied dry.

CAUTION: Some people have experienced allergic reactions when exposed to epoxy or cyanoacrylate glues. This is very rare. However, it is always important that such glues, and also paints, thinners, and solvents, be used with adequate ventilation to carry fumes away.

ABOUT BALSA WOOD:

We do our best to put as good a grade of balsa in our kits as the supply situation permits. The world-wide increase in demand for balsa has made it impossible to obtain as high an average quality as used to be the case.

Every piece of balsa supplied cannot be 100% perfect or kit prices would have to be greatly increased. Mineral stains or small knots do not affect wood strength. Even with the very best grades of balsa, there is a natural tendency for some sticks or sheets to immediately bow upon being cut off from a perfectly square block because of built-in stresses. In most cases, these can be bowed back into alignment during building. True up the edges of bowed sheets by trimming, using a metal straightedge to cut along.

IMPORTANT NOTE: Carefully align the wing and the stabilizer with each other when viewed from the front and from above.

WARNING-DANGER!

DO NOT FLY MODEL AIRPLANES NEAR ELECTRIC POWER LINES. INSTANT DEATH CAN RESULT FROM CONTACT WITH, OR FLYING TOO CLOSE TO, ELECTRIC POWER LINES.

READ INSTRUCTIONS AND STUDY PLANS CAREFULLY BEFORE BEGINNING CONSTRUCTION.

SIG®

SKYRAY 35

CL-25

DESIGNED AND DRAWN BY:

Mike Pratt

WING SPAN: 44 IN.
LENGTH: 33 IN.
ENGINES: .19 TO .35 CU. IN.

FUSELAGE CONSTRUCTION

NOTE: The balsa fuselage is cut oversize at the nose to avoid damage during shipment. The excess balsa will be trimmed away after the plywood fuselage doublers are glued in place.

STEP (1) The stock motor mount spacing of the Skyray 35 is for a Fox .35 Stunt. If you are going to use an engine other than the Fox .35 Stunt, you must adjust the motor mount spacing to fit your particular engine. Measure the width of the engine crankcase at the mounting lugs and adjust the spacing of the maple motor mounts by trimming away balsa from the fuselage as needed.

STEP (2) Apply a thin coat of slow-drying epoxy glue to the inside of one of the plywood doublers. Lay the balsa fuselage in place over it and epoxy glue the 3/8" x 1/2" maple motor mounts in place. Add the 3/8" x 1/2" x 6" balsa fill-in to the bottom of the lower maple motor mount. Apply a thin coat of epoxy to the inside of the other plywood doubler and lay it on the fuselage. Once you are sure that all of the parts are properly aligned, clamp or weight the assembly down and allow it to dry.

STEP (3) Trim away the excess balsa on the nose with a modeling knife and round all the corners of the fuselage with a sanding block, except for the area where the stabilizer is to be mounted.

STABILIZER & RUDDER ASSEMBLY

STEP (1) Slightly bevel the front edge of the rudder with a sanding block to achieve a 1/4" offset to the right (to the outside of the flying circle). Glue the fin and rudder pieces together.

STEP (2) Hold the stabilizer and elevator together with masking tape and round the tips of the elevator as shown on the plan. Round all the outside edges with a sanding block, then remove the masking tape and round the edges where the stabilizer and elevator are hinged.

BUILDER'S TIP: If you are going to cover your Skyray with an iron-on pre-colored covering material (like Sig Supercoat, etc.) it is best to cover the stabilizer and elevator before hinging them together. Refer to the section "COVERING WING AND TAIL SURFACES".

LIMIT OF LIABILITY

In use of our products, SIG Manufacturing Co.'s only obligation shall be to replace such quantity of the product proven to be defective. User shall determine the suitability of the product for his or her intended use and shall assume all risk and liability in connection therewith.

WARNING — DANGER!
Important: Read These Warnings:

Do Not fly control line or towline models within 300 feet of electric power lines. Instant death from electrocution can result from coming near them. Direct contact is not necessary.

A model airplane motor gets very hot and can cause serious burns. Do not touch the motor during or after

INSTALLING EASY HINGES

STEP (A) Place the stabilizer and elevator over the plans and mark the location of the hinges with a pencil. Using a No. 11 X-Acto blade (or similar) cut a slot approximately 1/2" in depth and slightly wider than the hinge at each of the marked hinge locations. After all of the slots have been cut, insert an Easy Hinge halfway into each slot of the stabilizer. Then carefully slide the elevator onto the other half of the hinges. You will find it easiest to slide the elevator onto the hinges at an angle, one at a time.

STEP (B) At this point the stabilizer and elevator are attached but not glued. Align the stabilizer and elevator and adjust the gap between them by flexing the elevator to its maximum deflection in both directions. For best control response, the gap should generally be as small as possible but big enough to allow the elevator to move to the maximum deflection that you will require.

STEP (C) Place three or four drops of any brand cyanoacrylate adhesive (thinnest variety) directly onto the Easy Hinge in the gap. You will notice that the glue is quickly wicked into the slot as it penetrates both the wood and the hinge. Continue this process, gluing the same side of all the hinges. Then turn the stabilizer and elevator over and repeat the gluing

FINAL ASSEMBLY

STEP (1) Carefully slide the wing through the cut-out in the fuselage. Extreme care must be taken to insure that the fuselage is centered on the wing. **NOTE:** It may be necessary to enlarge the wing cut-out hole slightly to fit the wing. Epoxy the wing to the fuselage. While the glue is drying make sure that the wing is square to the fuselage when viewed from above and from behind.

STEP (2) Drill out the bottom hole of the nylon control horn with a #41 drill bit to accept the 3/32" music wire pushrod. Attach the nylon control horn to the elevator with the two #2 sheet metal screws that are provided.

STEP (3) Temporarily pin the stabilizer/elevator assembly in place but do not glue it at this time. Insert the pushrod into the nylon control horn and solder two #3 flat washers onto the pushrod, again using paper spacers to protect the nylon control horn from the heat.

STEP (4) With the bellcrank in the neutral position (parallel to the fuselage), slide the stabilizer/elevator assembly forward or backward until the elevator is level and epoxy glue the assembly in place at that point. Before the epoxy glue cures, double check the alignment of the stabilizer/elevator assembly by viewing it from the front of the model. It

FINISHING THE FUSELAGE

All of the Skyray 35 proto-types had their wings and tail surfaces covered with Sig Supercoat iron-on plastic covering (Skyblue SCC-414). The fuselages were painted a matching color with Sig Supercoat Color Dope (Skyblue SD-032). For a great looking model in the shortest amount of time, this procedure is highly recommended.

Sand the entire model very carefully with 220 grit Tri-M-Ite sandpaper, and add fillets made from Sig Epoxolite (EP-001). Fillets not only improve the looks of the model, they greatly increase the strength of the fuselage-to-wing joint. Fill-in any dents or dings in the structure at this time.

Brush on three coats of Sig Lite-Coat clear low shrink dope to the fuselage and to about a 1" band of the center sheeting on both sides of the fuselage. Lightly sand between each coat with 220 grit Tri-M-Ite finishing sandpaper. Brush on two coats of Sig Sanding Sealer, again sanding between each coat with 220 sandpaper. Then brush or spray on one more coat of Sig Lite-Coat clear to seal off the sanding sealer and start a gloss build-up. Next, spray or brush on your "base color" coats. When completely dry, spray on two top coats of Sig Lite-Coat clear. The top coats not only add gloss to your model, they help protect your finish as well.

COVERING WING & TAIL SURFACES

The following instructions are intended for Sig Supercoat Covering. However, the basic techniques for applying iron-on plastic coverings of any brand are pretty much the same and the following hints should be helpful.

NOTE: If you choose another brand of covering material, be sure to read the manufacturer's directions that come with the material. Follow their instructions closely when applying the material, as different brands of coverings can have slightly different handling characteristics and application temperatures.

WING

You should start by covering the bottom halves of the wing first and then the top halves. This leaves the overlapping seam on the bottom where it is less visible. Cut the covering to size, allowing approximately one inch of excess around the edges.

Remove the plastic backing from the covering material and lay the adhesive side against the wing. **NOTE:** The covering material should

With a sealing iron or a heat gun, start shrinking the plastic covering material. Work slowly by rotating from the top of the wing to the bottom, shrinking only one or two rib bays at a time. Sight along the rear of the trailing edge of the wing occasionally to make sure that you are not adding any warps or twists to the wing.

TAIL SURFACES

The structure that is to be covered must be clean, dry, and dust free. Wipe the entire surface with a tack rag or a rag dampened with alcohol to remove all excess dust.

Cut out a piece of covering material slightly larger than the stabilizer. Remove the plastic backing from the covering and lay the adhesive side of the material in place on the stab. Using your hot sealing iron (set at 200⁰F. for Sig Supercoat) seal down the covering, working from the middle of the stabilizer out towards the edges. Work the covering material completely around the edges of the stabilizer and trim off the excess with a sharp razor blade. Repeat the above procedure for the opposite side of the stabilizer.

LANDING GEAR

Install the landing gear on the inboard side of the fuselage by drilling three 1/8" holes in the locations shown on the plan. Bolt the gear in place with the nylon landing gear clip and two 4-40 x 1" bolts and blind nuts. Slide the wheel over the axle and secure it in place with a 1/8" wheel collar.

FUEL TANK

The fuel tank that is shown on the plan is a Sullivan 4 oz. slant fuel tank (#437). These clunk type fuel tanks are more commonly found in R/C models, however, they are easily adapted to control-line models and will work extremely well when set-up properly.

Only two pieces of brass tubing are necessary for this type of tank. (1) A fuel vent/overflow tube, (2) A fuel pick-up tube. The fuel vent/overflow tube must be located in the top inboard corner of the fuel tank (Refer to the fuel tank drawings on the plan). When viewed from the side, the fuel pick-up tube must be lined up directly with the center of the engine venturi. Assemble the fuel tank as per the manufacturer's instructions making sure the clunk is located in the lower outboard corner of the fuel

ENGINE

Mount the engine in place with four 4-40 x 1" bolts and blind nuts. Don't forget to add a #3 flat washer under each of the front engine mounting lugs for the required amount of engine offset.

FLYING

The Skyray makes an excellent stunt trainer when equipped with a .35 engine. It's more than capable of performing the entire AMA stunt pattern with a little practice.

Fly your Skyray 35 on .015 x 60" braided steel control lines (SH-457) and use a sturdy control handle. DO NOT USE 1/2A size flying equipment. Attach the control lines to the model's leadouts and to the control handle with AMA-approved control line clips (Sig SH-443). Adjust the control handle so that the elevators are in the neutral (level) position when the control handle is straight up and down.

Be sure to pick a flying site that is free of obstacles and far from houses and buildings.

CAUTION: NEVER FLY NEAR ELECTRIC POWER LINES. INSTANT